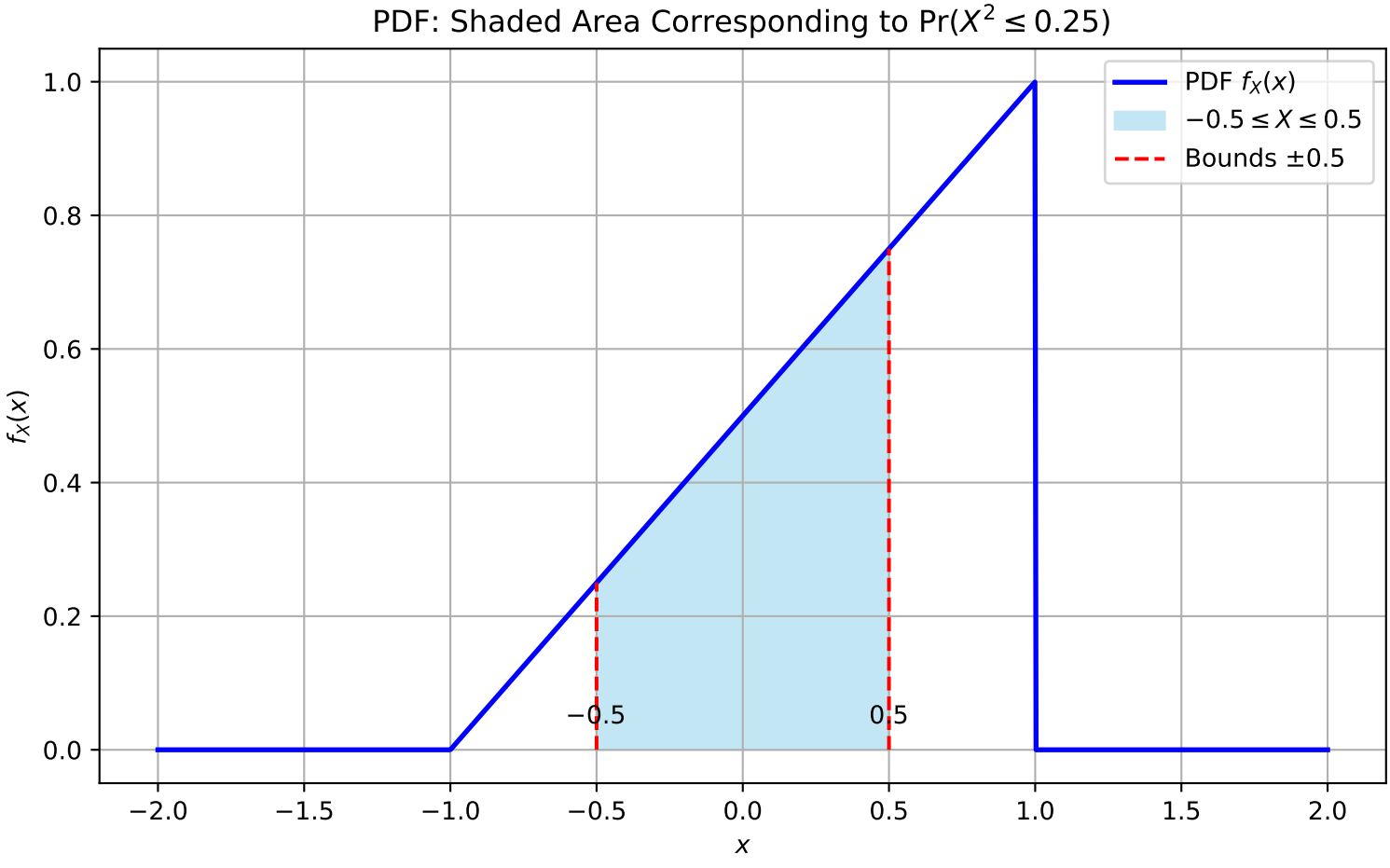
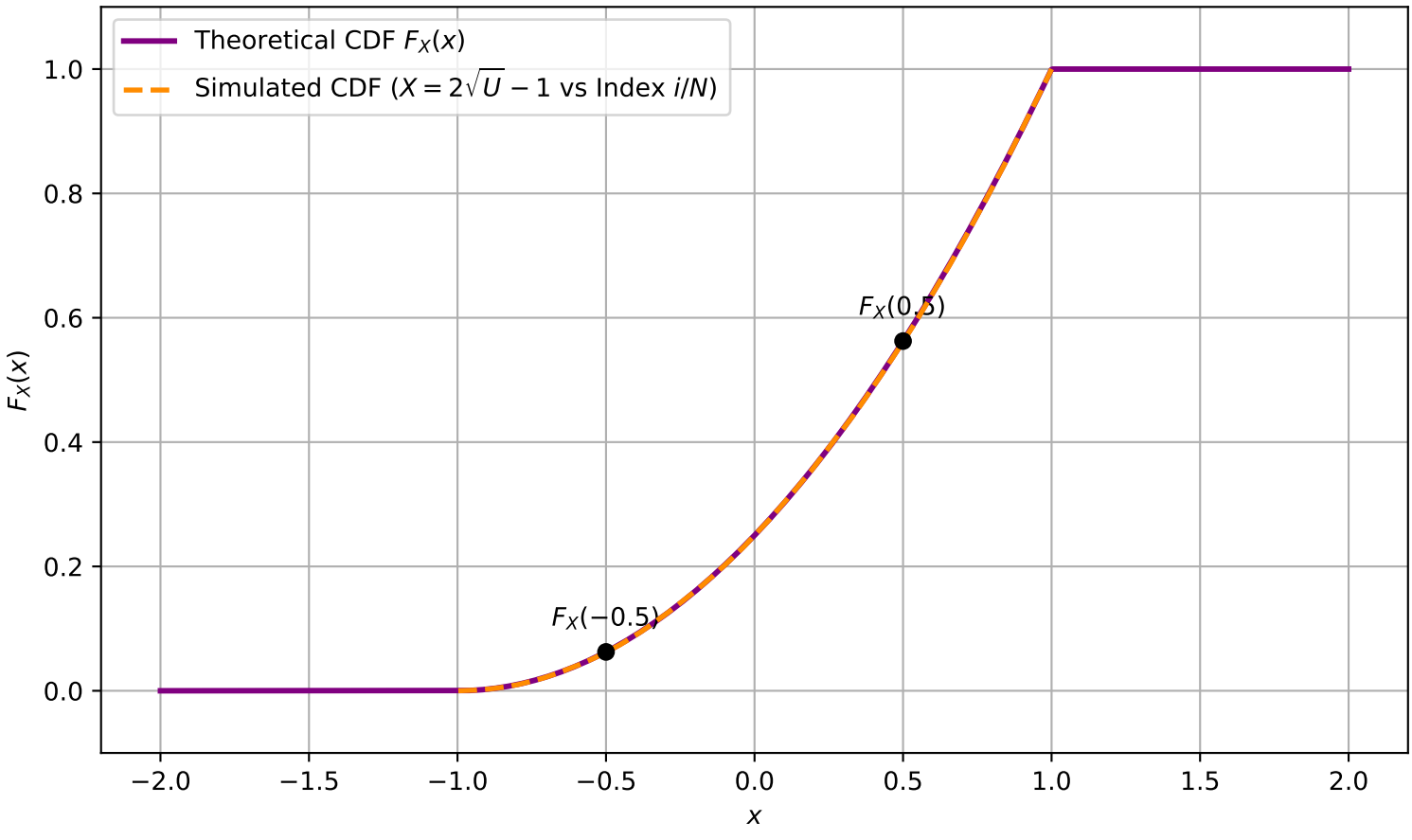

PDF: Shaded Area Corresponding to $\Pr(X^2 \leq 0.25)$
PDF $f_X(x)$
$-0.5 \leq X \leq 0.5$
Bounds ± 0.5
-0.5
0.5
$f_X(x)$
x


CDF: $\Pr(X^2 \leq 0.25) = F_X(0.5) - F_X(-0.5) = 0.5$
Theoretical CDF $F_X(x)$
Simulated CDF ($X = 2\sqrt{U} - 1$ vs Index i/N)
$F_X(0.5)$
$F_X(-0.5)$
$F_X(x)$
x